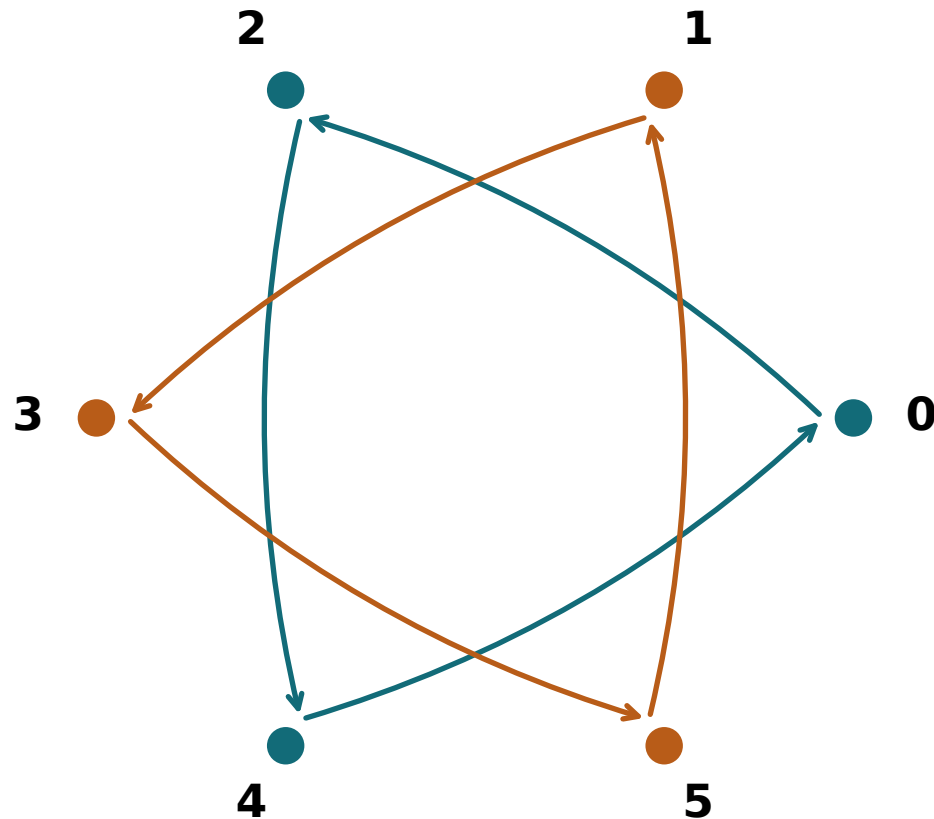


Orbit coordinates: six points, two inequivalent representations



Exact points: $\exp(i\pi j/3)$, $j = 0, \dots, 5$

The action is $h \cdot j = j + 2h \pmod{6}$.

$G = \mathbb{Z}/3\mathbb{Z}$; uniform probability on all six points.

It is free, with orbits $\{0, 2, 4\}$ and $\{1, 3, 5\}$.

For $p_j(f)$, coordinate h reads $f(j + 2h)$.

Move j to $j+2$: $V_1 \delta h = \delta h - 1 \pmod{3}$.

$h = 0 \rightarrow 2 \quad h = 1 \rightarrow 0 \quad h = 2 \rightarrow 1$

The right shift commutes with all left shifts.

Same orbit: unitarily equivalent irreducibles.

Different orbits: all intertwiner entries vanish.

The algebra is $M_3 \oplus M_3$; its center is $\mathbb{C}^2$.

The actual regular Hilbert space has dimension 18.

OS0-OS2 and WK0: all six support-point states are pure; the diagonal integral is orthogonal.

This finite model is outside the countably infinite, free ergodic type-classification hypotheses. Here every density d_g equals 1.